File-System Backups

1 Basic Idea

Definition

File-system backup means copying file-system data to separate storage so lost, damaged, or accidentally deleted files can be restored.

Why It Matters

Replacing a computer is usually easier than reconstructing lost programs, documents, databases, records, and project files.

2 Two Main Reasons

Reason	Meaning
Recover from disaster	Disk crash, fire, flood, theft, or system failure.
Recover from stupidity	User accidentally deletes or overwrites files.

3 Backup Scope

Usually back up	Usually skip
User files	Reinstallable binaries.
Configuration files	Temporary directories.
Databases and project data	Device files such as `/dev`.
Important system state	Paging, hibernation, and other rebuildable internal files.

Device File Risk

Backing up special device files can be dangerous. A backup program may hang if it tries to read a device file as ordinary data.

4 Full and Incremental Backups

Type	What it saves	Trade-off
Full backup	Everything selected for backup.	Easy restore, slow backup.
Simple incremental	Files changed since the last full backup.	Faster backup, restore needs full plus incrementals.
Advanced incremental	Files changed since their last backup.	Saves time, but recovery order is more complex.

Restore Rule

Restore the latest full backup first, then apply incremental backups in order until the desired date is reached.

5 Backup Problems

Issue	Point
Compression	Saves space, but one bad tape spot may break decompression.
Active file system	Files may change while backup is running, causing inconsistency.
Snapshots	Freeze a logical view quickly; later writes use copy-on-write.
Security	Backup media must be protected like the original data.
Off-site copies	Protect against fire or local disaster, but add another site to secure.

6 Physical vs. Logical Dump

Dump type	Main idea	Limitation
Physical dump	Copy disk blocks in order, starting at block 0.	Hard to skip directories, do incrementals, or restore one file.
Logical dump	Start at selected directories and copy named files/directories.	More complex algorithm.

Practical Result

Physical dumping is simple and fast. Logical dumping is more useful for normal backups because it can restore selected files and support incremental backups.

7 Physical Dump Notes

- Do not waste time copying unused disk blocks.
- If unused blocks are skipped, the backup must record original block numbers.
- Bad blocks must be avoided, or the backup may repeatedly hit read errors.
- System-specific files, such as paging or hibernation files, may be skipped.

8 Logical Dump Example

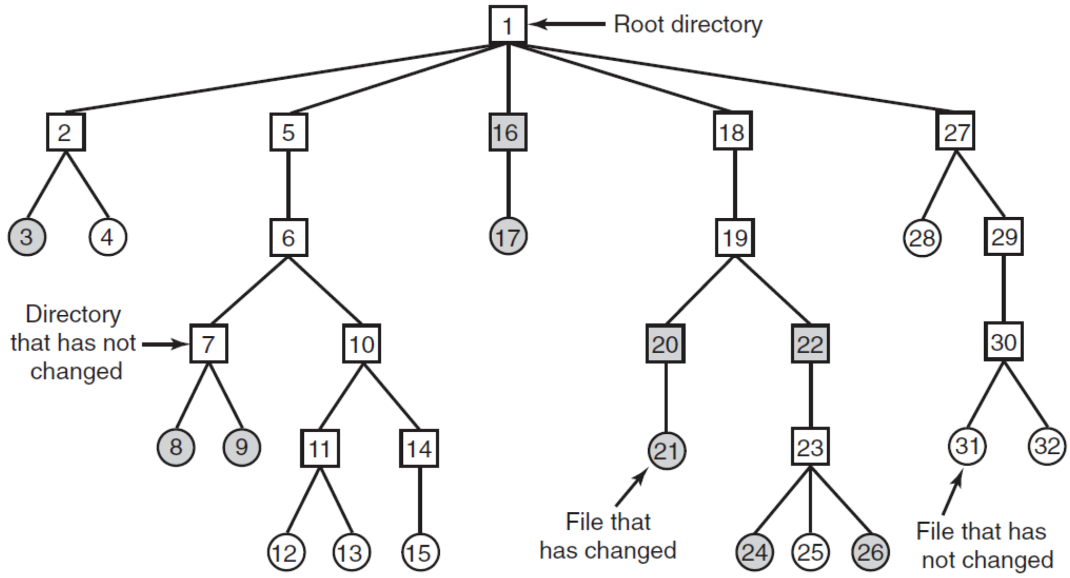

Figure 4-25. A file system to be dumped. The squares are directories and the circles are files. The shaded items have been modified since the last dump. Each directory and file is labeled by its i-node number.

Figure Meaning

Squares are directories, circles are files, and shaded items have changed since the base backup date.

Directory Rule

Dump modified files plus every directory needed to reach them, even if those directories did not change.

9 Why Unchanged Parent Directories Are Dumped

1. They allow the dump to be restored onto a fresh empty file system.
2. They allow one deleted file to be restored later with its full directory path.
3. They preserve directory attributes such as owner, mode, and timestamps.

10 Logical Dump Bitmap Algorithm

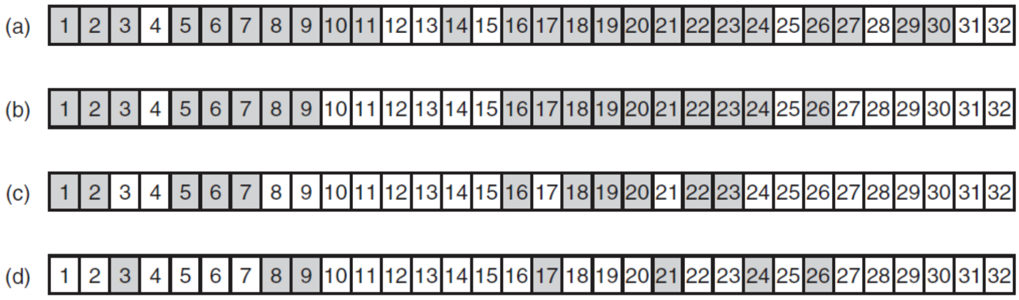

Figure 4-26. Bitmaps used by the logical dumping algorithm.

Phase	Action
1	Mark all modified files and all directories.
2	Unmark directories that contain no modified file or needed directory below them.
3	Dump marked directories in i-node order, with attributes.
4	Dump marked files in i-node order, with attributes.

11 Restoring

1. Create an empty file system.
2. Restore directories first to build the skeleton tree.
3. Restore file contents.
4. Apply later incremental dumps in sequence.
5. Rebuild the free-block list from the files that now exist.

12 Tricky Cases

Case	Required handling
Hard links	Restore file data once, then recreate all directory names pointing to it.
Sparse files	Do not dump holes; restore holes as holes, not blocks full of zeros.
Special files	Do not dump device files, named pipes, or other non-regular files as data.
Free list	Reconstruct after restore; it is not an ordinary file.

Sparse File Warning

If holes are restored as zero-filled blocks, a small sparse file can become enormous.

13 Final Summary

Summary

Backups protect against disasters and accidental deletion. Physical dumps are simple but inflexible; logical dumps are more useful because they support selected directories, incremental backups, and single-file restore.